

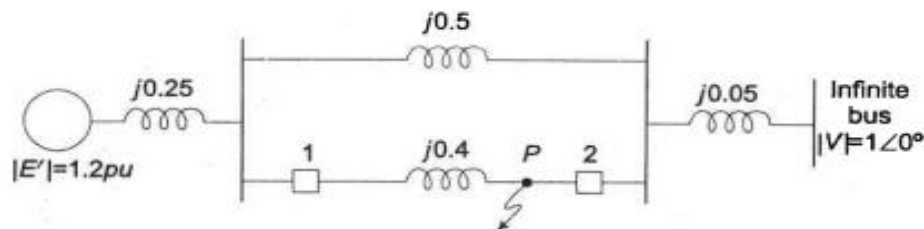
Module 2- Power system stability

❖ Theory based conceptual question

- Find out frequency of oscillation when a power system experiences small disturbances in the context of steady state stability? [10]
- Using swing equation for multi-machine stability analysis derive the expression for equivalent inertia constant when the two machine operating under following condition
 - Coherent operation
 - Non coherent operation
 [5+5]

❖ Comprehensive numerical Question

- A 50 Hz four pole turbo-generator rated 20 MVA , 13.2kV has an inertia constant of $H=9.0$ kW-sec/kVA.
 - Determine the acceleration if the input less the rotational losses is 25000HP and the electric power developed is 15000kW.
 - If the acceleration computed for the generator is constant for a period of 15 cycles; determine the change in torque angle in that period and the rpm at the end of 15 cycles. [Assume that the generator is synchronized with a large system and has no accelerating torque before the 15 cycle period begins.] [4+6]
- A generator delivers 1 pu to an infinite bus through purely reactive network. The maximum power that could be delivered by the generator is 2 pu. A 3-phase fault occurs at the terminal of generator which reduces the electrical output to zero. The fault is cleared after t_c second. The original network is then restored. The maximum rotor angle is 110° . What will be the rotor angle in electrical degree at $t=t_c$. [5]
- A synchronous generator is connected with IBB through a lossless double circuit transmission line. The generator is delivering 1 pu of power at an angle of 30° . Sudden fault reduces the peak power to 0.5 pu. After clearance of fault, the peak power that can be transmitted becomes 1.5 pu. Find $\delta_{cr} = ?$ [5]
- Follow the following network. [10]



Find the critical clearing angle for clearing the fault with simultaneous opening of the breakers 1 and 2. The reactance values of various components are indicated on the diagram. The generator is delivering 1.0 pu power at the instant preceding the fault.

Assignment 1